Module 7: Lesson 2

# Advanced topics in Deep Learning for Finance: Application of Walk Forward

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# Outline

- The Walk Forward method in practice
- Application to hourly Bitcoin data

# Walk Forward

In Lesson 1, we described the Walk Forward method as a tool to increase the robustness of our evaluations of trading strategies.

With a Walk Forward, we can perform backtests using more information than that of the last chunk of data from a standard train/test split, and we reduce the bias of our evaluations toward the most recent part of the data.

In Lesson 2, we take this method into practice, using a non-anchored Walk Forward (fixed-size rolling window).

# Walk Forward: Application to Bitcoin

In our illustration of the Walk Forward, we exploit high-frequency Bitcoin price data.

The dataset covers hourly information, provided transactions have occurred, in the 2013–2022 period.

To ease the computational requirements, we only use information from the 2020–2022 period.

| | date | volume | trades | Ret | year |
|---|---|---|---|---|---|
| **43802** | 2020-01-01 00:00:00 | 70.761775 | 267 | -0.001758 | 2020 |
| **43803** | 2020-01-01 01:00:00 | 224.921206 | 528 | 0.005408 | 2020 |
| **43804** | 2020-01-01 02:00:00 | 138.910528 | 336 | 0.003906 | 2020 |
| **43805** | 2020-01-01 03:00:00 | 97.687456 | 240 | -0.002908 | 2020 |
| **43806** | 2020-01-01 04:00:00 | 25.305555 | 131 | -0.001236 | 2020 |
| **...** | ... | ... | ... | ... | ... |
| **67886** | 2022-09-30 19:00:00 | 290.023381 | 1947 | -0.008941 | 2022 |
| **67887** | 2022-09-30 20:00:00 | 759.074836 | 1648 | -0.003939 | 2022 |
| **67888** | 2022-09-30 21:00:00 | 564.894247 | 1245 | -0.003090 | 2022 |
| **67889** | 2022-09-30 22:00:00 | 90.626135 | 1978 | 0.001379 | 2022 |
| **67890** | 2022-09-30 23:00:00 | 57.986112 | 2466 | 0.001965 | 2022 |

24089 rows × 5 columns

# Walk Forward: Deep Learning Architecture

We employ an MLP architecture for a binary classification problem.

- The label to predict is the direction of the cumulative Bitcoin return over the next 4-hour interval.
- The input features are the lags of the hourly Bitcoin returns over the previous 15 days.

The MLP features three fully connected hidden layers with, respectively, 85, 55, and 30 hidden units.

- Each hidden layer features a ReLU activation function and a sigmoid function in the output layer.
- For regularization, we introduce Dropout after all the hidden layers and early stopping.

# Walk Forward: Implementation

In the Walk Forward, we train the MLP separately for successive training samples of fixed size, using the procedure outlined in Lesson 1.

We set the size of the training window to $\mathcal{T}_1$=5,000 hours of data, while we set the test window to $\mathcal{T}_2$=1,000 hours.

The procedure involves nesting the usual lines of code that involve training a model in TensorFlow into a for loop of the form:

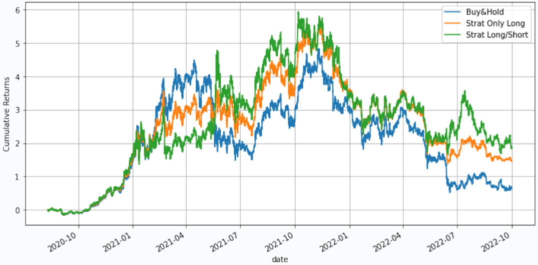
```
n_train = 5000
n_test = 1000

for i in range(0, len(y)-n_train, n_test):
  X_train, X_test = X[i:i+(n_train)], X[i+n_train:i+n_train+n_test]
  y_train, y_test = y[i:i+(n_train)], y[i+n_train:i+n_train+n_test]
```

We store in a pandas data frame the prediction results for each test observation.

# Walk Forward: Results

Once we have trained the model for all the training windows, we can obtain backtest statistics for a larger set of data than the usual single training/testing split.

- We can obtain out-of-sample predictions for all the datasets, except for the initial training sample.




Disclaimer: The results above can vary due to the randomness of the training process.

# Summary of Lesson 2

In Lesson 2, we have:

- Revisited the Walk Forward method.
- Implemented the Walk Forward using high-frequency data.

⇒ **References for this lesson:**

Gray, Chad. "Stock Prediction with ML: Walk-forward Modeling." *The Alpha Scientist*, July 2018.

**TO DO NEXT**: Please go to the Jupyter Notebook associated with Lesson 2.

In the next lesson, we will describe a more advanced method to obtain robust backtests: the Combinatorial Purged Cross-Validation.

See you there!